

MARWADI UNIVERSITY

DEPARTMENT OF AI, ML

DAA (01AI0506) — Lab Manual

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PRACTICAL - 7

Implementation of Making Change Problem using Dynamic Programming

Aim of the Practical:

To find the minimum number of coins required to make a target amount using dynamic programming and analyze its complexity.

Algorithm:

- 1 Initialize $dp[0] = 0$ and $dp[1...A] = \infty$.
- 2 For each amount x from 1 to A , try every coin denomination.
- 3 If $coin \leq x$ and $dp[x-coin]$ is finite, update $dp[x] = \min(dp[x], dp[x-coin] + 1)$.
- 4 Return $dp[A]$, or -1 if it is still infinite.

Program:

```
#include <iostream>
#include <vector>
#include <climits>
#include <algorithm>
using namespace std;

int minCoins(const vector<int>& coins, int amount) {
    vector<int> dp(amount + 1, INT_MAX);
    dp[0] = 0;

    for (int x = 1; x <= amount; x++) {
        for (int coin : coins) {
            if (coin <= x && dp[x - coin] != INT_MAX) {
                dp[x] = min(dp[x], dp[x - coin] + 1);
            }
        }
    }

    return dp[amount] == INT_MAX ? -1 : dp[amount];
}

int main() {
    vector<int> coins = {1, 3, 4};
    int amount = 6;
    cout << minCoins(coins, amount);
}
```

Output:

2

Step-by-Step Analysis:

- $dp[0] = 0$
- $dp[1] = 1$ using {1}
- $dp[2] = 2$ using {1,1}

- $dp[3] = 1$ using $\{3\}$
- $dp[4] = 1$ using $\{4\}$
- $dp[5] = 2$ using $\{4,1\}$
- $dp[6] = 2$ using $\{3,3\}$

Time Analysis:

- Time Complexity: $\Theta(Am)$.
- Space Complexity: $\Theta(A)$.

Result:

The Making Change problem was solved successfully using dynamic programming, and the minimum number of coins for amount 6 is 2.